The Area of an SAS Triangle

The area of a triangle is equal to the length of one side times the length of another side times the sine of the measure of the angle between the two sides divided by two.

$$\text{Area} = \frac{1}{2}ab\sin C \qquad \text{Area} = \frac{1}{2}bc\sin A$$

$$\text{Area} = \frac{1}{2}ac\sin B$$

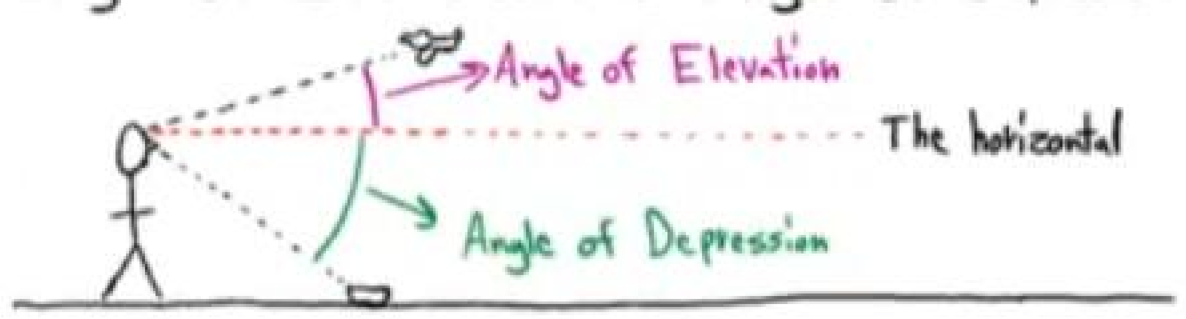
Heron's Formula

If a, b, and c are the side lengths of a triangle, then the area of the triangle is equal to the expression below.

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

where $s = \frac{a+b+c}{2}$

Angle of Elevation and Angle of Depression



Converting from Degrees to Radians

degree measure $\times \frac{\pi \text{ radians}}{180 \text{ degrees}}$ = radian measure

Converting from Radians to Degrees

Radian measure $\times \frac{180 \text{ degrees}}{\pi \text{ radians}}$ = degree measure

Arc Length

$\boxed{s = \theta r}$ *This formula can be applied with radian units only

The Area of a Sector

$\boxed{A = \frac{\theta}{2}r^2}$ *This formula can be applied with radian units only.

If an object moves s units on a circular path in t units of time, then V is defined as the linear velocity according to the equation below.

$$V = \frac{s}{t}$$

If an object on a circle rotates θ angle units in t units of time, then ω is defined as the angular velocity according to the equation below.

$$\omega = \frac{\theta}{t}$$

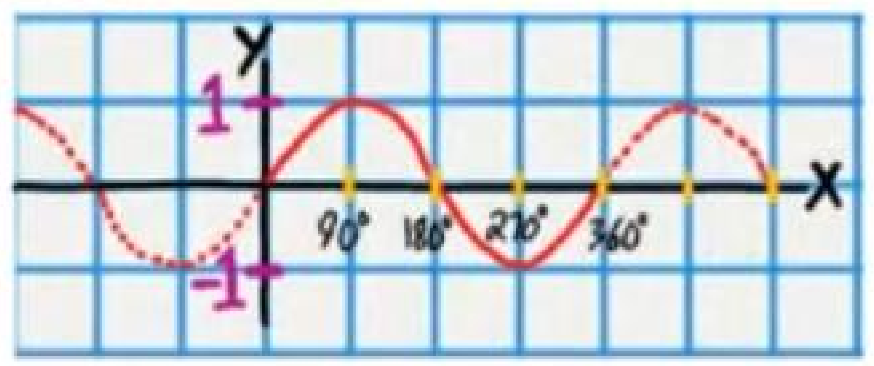
$\boxed{V = r\omega}$ *This formula can be applied with radian units only

Revolution: A complete rotation.

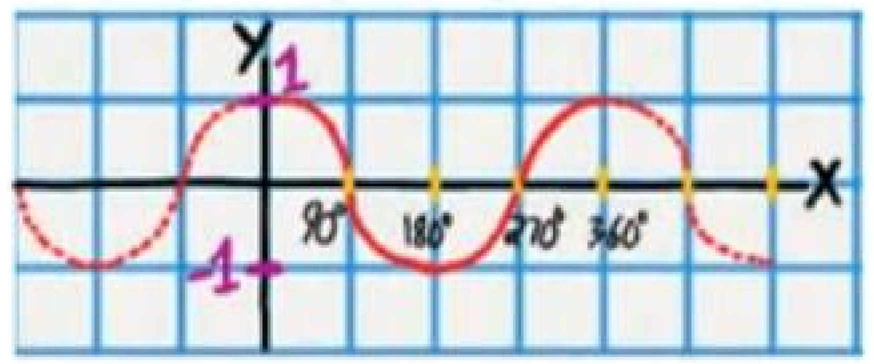
Revolutions per minute = rpm

1 Revolution = 2π radians = 360°

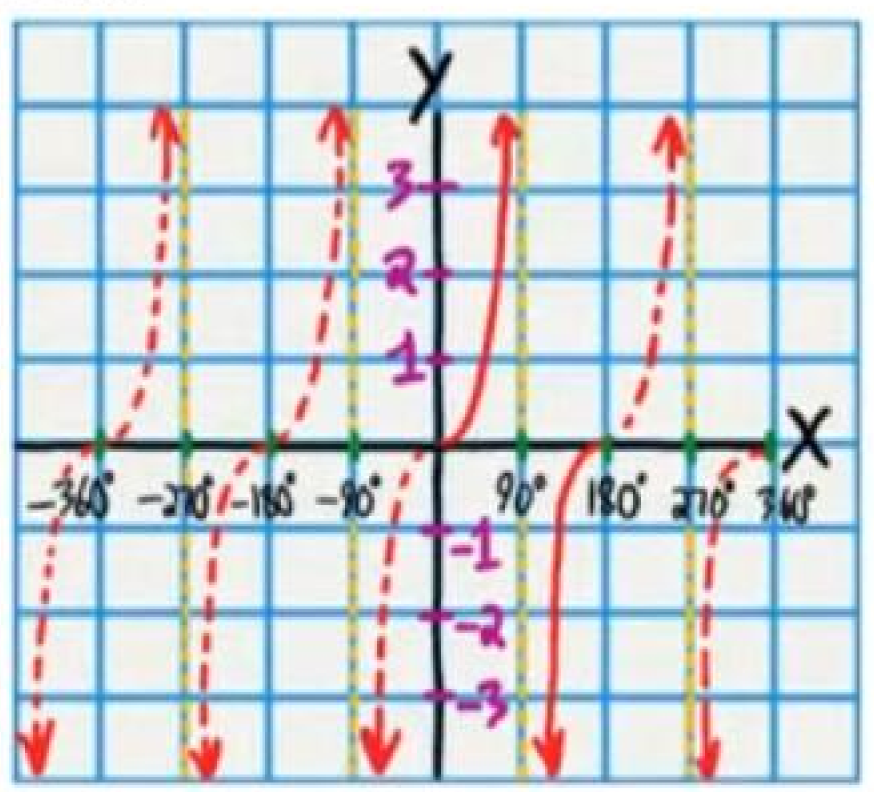
y = sinx



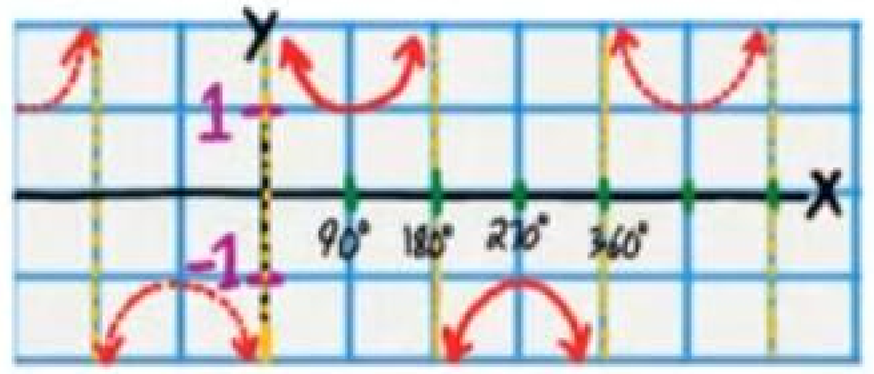
y = cosx



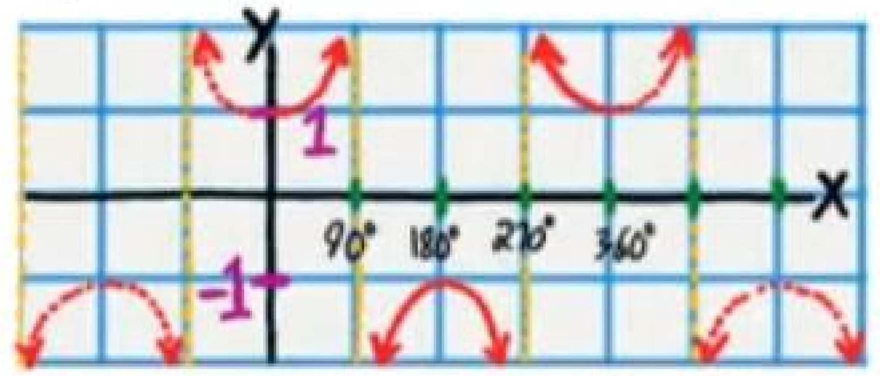
y = tanx



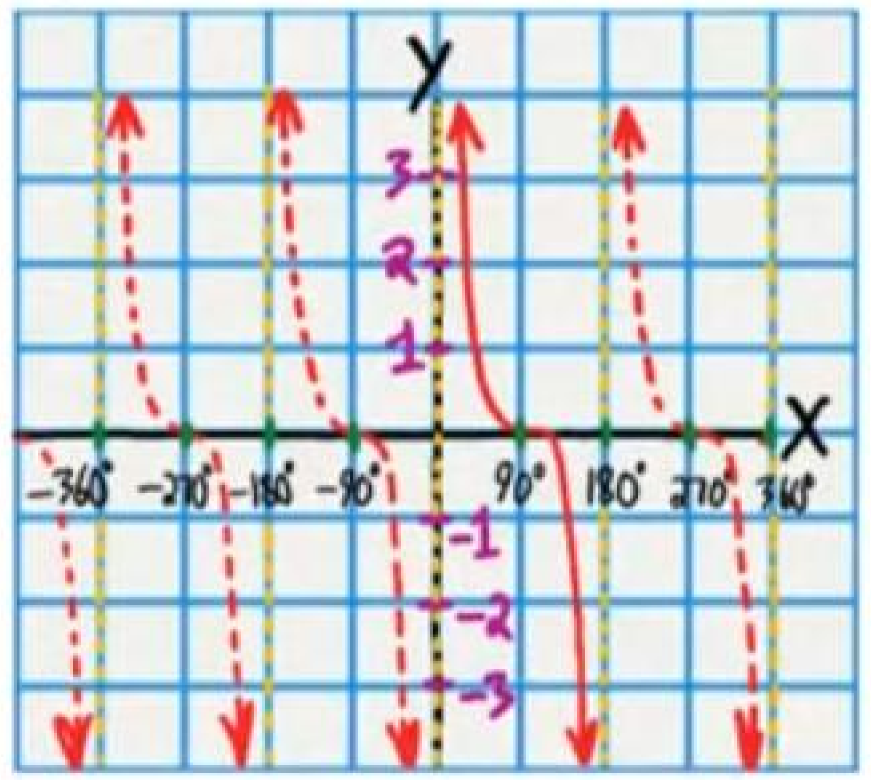
y = cscx

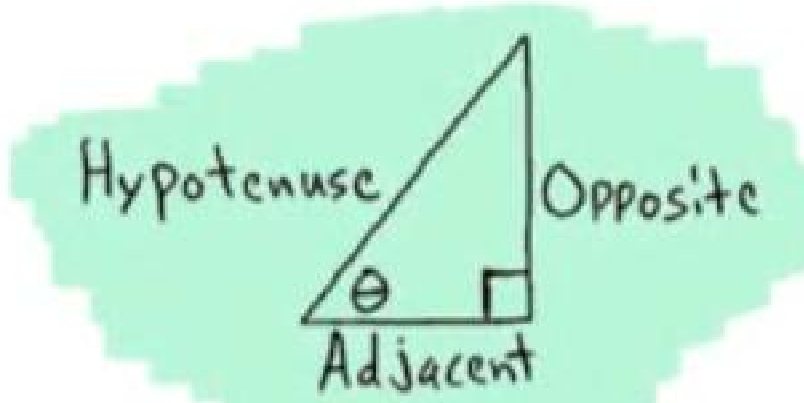


y = secx



y = cotx



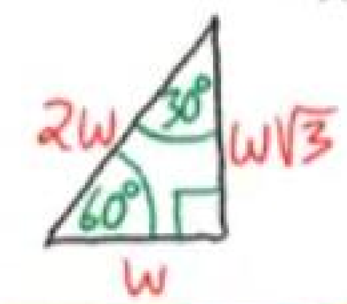


$$\sin\theta = \frac{\text{Opp}}{\text{Hyp}} \qquad \csc\theta = \frac{\text{Hyp}}{\text{Opp}}$$

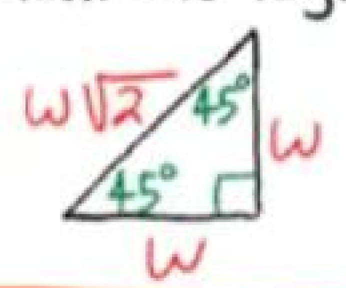
$$\cos\theta = \frac{\text{Adj}}{\text{Hyp}} \qquad \sec\theta = \frac{\text{Hyp}}{\text{Adj}}$$

$$\tan\theta = \frac{\text{Opp}}{\text{Adj}} \qquad \cot\theta = \frac{\text{Adj}}{\text{Opp}}$$

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is ω, then the length of the hypotenuse is 2ω and the length of the side opposite the 60° angle is $\omega\sqrt{3}$.

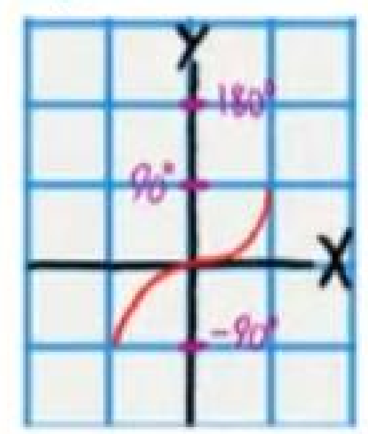


45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90° and the length of the two sides opposite the 45° angles are ω, then the length of the hypotenuse is $\omega\sqrt{2}$.



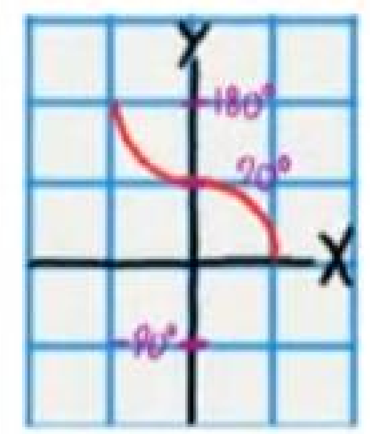
$y = \sin^{-1}x$

Range: $-90° \le y \le 90°$



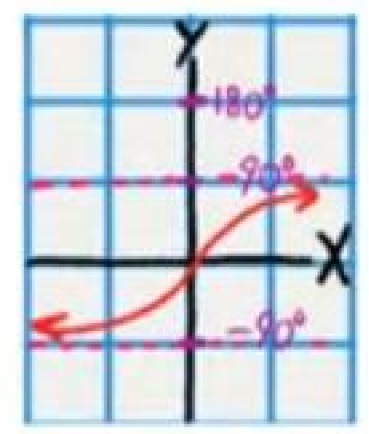
$y = \cos^{-1}x$

$0° \le y \le 180°$



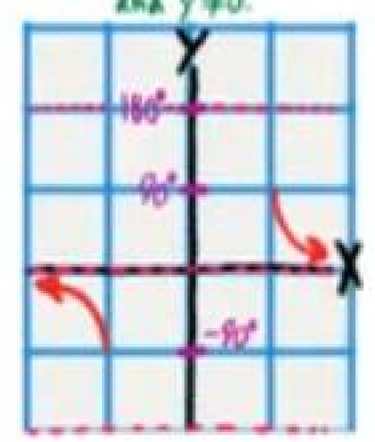
$y = \tan^{-1}x$

$-90° < y < 90°$



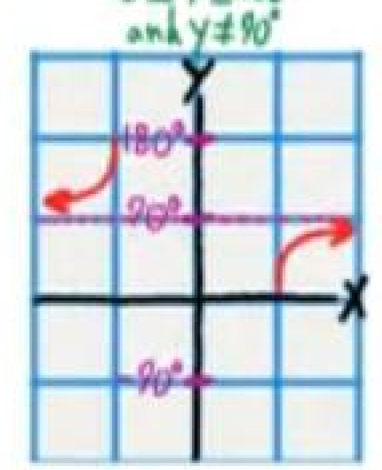
$y = \csc^{-1}x$

Range: $-90° \le y \le 90°$ and $y \ne 0$.



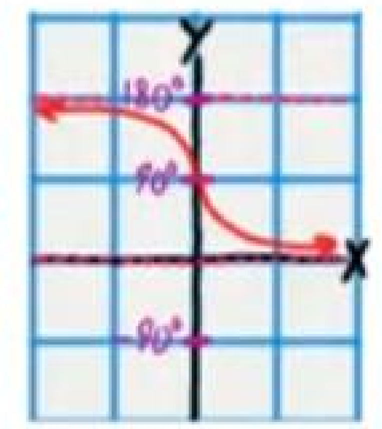
$y = \sec^{-1}x$

$0° \le y \le 180°$ and $y \ne 90°$



$y = \cot^{-1}x$

$0° < y < 180°$



Name
Date
Course

Practice Test IV (Version 1)

I) Write your name, the date, and the course title (i.e. Trigonometry).
II) Show all work.
III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) You will need a calculator for this test. When pi is required, use the π button on your calculator. Do not use 3.14.
V) Each problem is worth four points for a total of 100 points.

① i) Convert the following angle measure to radian units.

$$50^\circ \times \frac{\pi}{180^\circ} = \frac{50\pi}{180} = \boxed{\frac{5\pi}{18}}$$

ii) Convert the following angle measure to degree units.

$$6\pi \times \frac{180^\circ}{\pi} = \frac{1{,}080^\circ\pi}{\pi} = \boxed{1{,}080^\circ}$$

② Find the values of the following expressions without a calculator.

i) $\csc\left(\frac{\pi}{2}\right)$

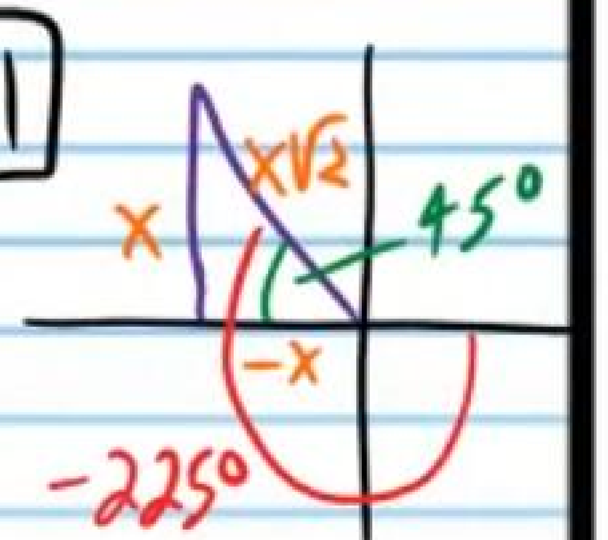
$$\frac{\pi}{2} \times \frac{180^\circ}{\pi} = 90^\circ$$

$$\csc 90^\circ = \boxed{1}$$

ii) $\tan\left(-\frac{5\pi}{4}\right)$

$$-\frac{5\pi}{4} \times \frac{180^\circ}{\pi} = -\frac{900\pi}{4\pi} = -225^\circ$$

$$\tan(-225^\circ) = \boxed{-1}$$



Find the value of the expression using a calculator. Round to the nearest thousandth.

③ $\cos\left(\frac{\pi}{5}\right) \approx \boxed{0.809}$

④ Find θ if $0 \le \theta \le 2\pi$. Write your answer in radian units.

$$\sec\theta - 9 = 7\sec\theta - 3$$

$$-9 = 6\sec\theta - 3$$

$$-6 = 6\sec\theta$$

$$-1 = \sec\theta$$

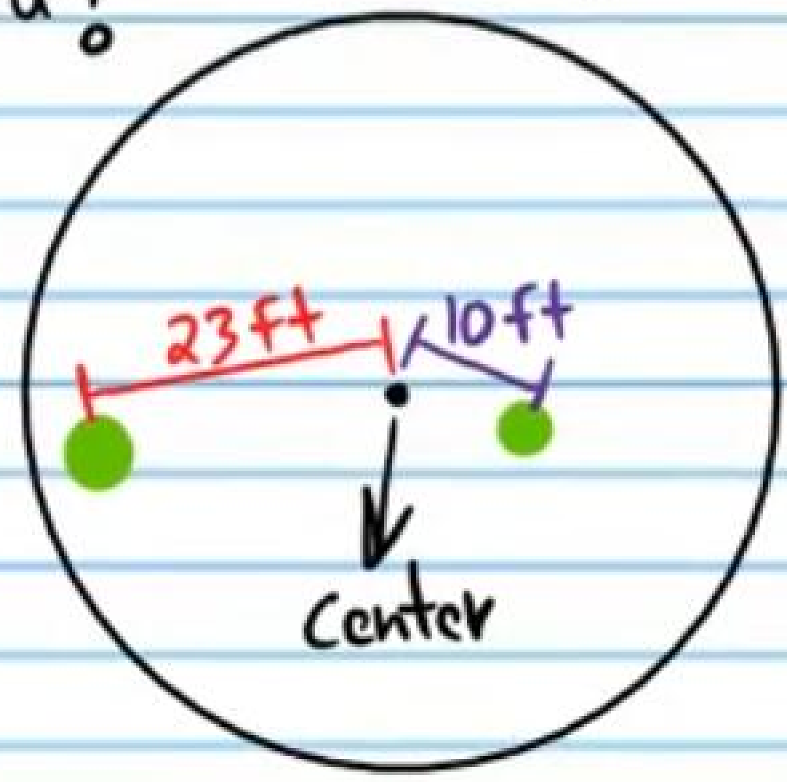
$$\theta = 180^\circ$$

$$\theta = 180^\circ \times \frac{\pi}{180^\circ} = \boxed{\pi}$$

⑤ Find the area of a sector of a circle with radius 7 miles and central angle $2\pi/5$.

$$A = \frac{\theta}{2}r^2 = \frac{2\pi/5}{2}(7^2) = \frac{2\pi}{10}(49) = \frac{\pi}{5}(49) = \boxed{\frac{49\pi}{5}\ \text{mi}^2}$$

⑥ A flying saucer (i.e. a circular unidentified flying object) rotates as it hovers (floats without moving) in the air. A little green man sits 23 feet from the center of the saucer and moves 8 feet per second. What is the angular velocity of the saucer? A second green man sits 10 feet from the center of the saucer? How many feet does he move per second?



I) $V_1 = r_1 \omega$

$8 = (23)\omega$

$\omega = \boxed{\frac{8}{23} \text{ rad/s}}$

II) $V_2 = r_2 \omega = 10\left(\frac{8}{23}\right) = \frac{80}{23} = \boxed{3\frac{11}{23} \text{ ft/s}}$

⑦ Alexandria runs on a circular track at a constant pace. She runs 4 laps in 27 minutes. If the radius of the track is 31 meters, how many meters does she run per second?

I) $\omega = \frac{4 \text{ laps}}{27 \text{ min}} = \frac{4 \text{ rev}}{27 \text{ min}}$

II) $\omega = \frac{4 \text{ rev}}{27 \text{ min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{8\pi \text{ rad}}{1620 \text{ s}} = \frac{2\pi}{405} \text{ rad/s}$

III) $V = r\omega = (31)\left(\frac{2\pi}{405}\right) = \boxed{\frac{62\pi}{405} \text{ m/s}}$

⑧ A record player spins at 45 rpm. If the player spins a record with a diameter of 24 inches, how many inches does a point on the outer edge move per minute?

I) $45 \text{ rpm} = \dfrac{45 \text{ rev}}{1 \text{ min}} \times \dfrac{2\pi \text{ rad}}{1 \text{ rev}} = 90\pi \text{ rad}/\text{min}$

II) $r = \dfrac{D}{2} = \dfrac{24}{2} = 12 \text{ in}$

III) $v = r\omega = (12)(90\pi) = \boxed{1{,}080\pi \text{ in}/\text{min}}$

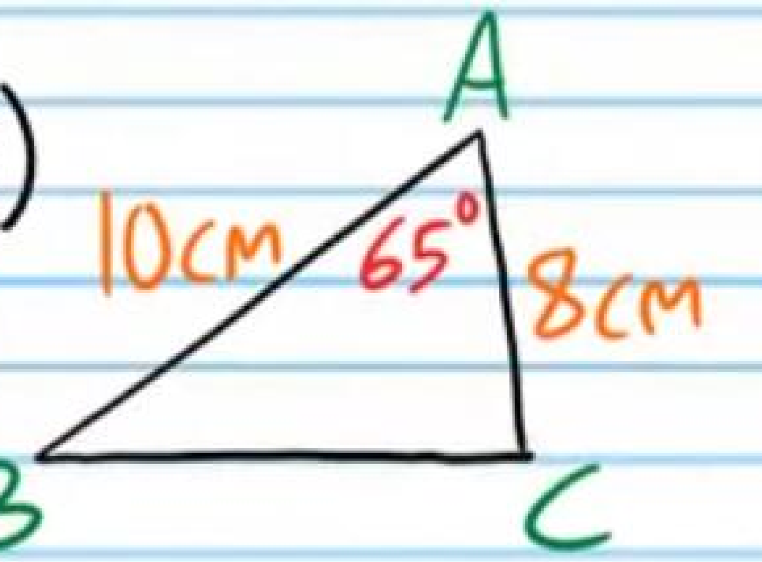
⑨ A merry-go-around completes one rotation in 1.2 seconds. If a child rides 2 meters from the center, how many meters does he move per second? Round to the nearest hundredth.

I) $\omega = \dfrac{1 \text{ rot}}{1.2 s} = \dfrac{1 \text{ rev}}{1.2 s} \times \dfrac{2\pi \text{ rad}}{1 \text{ rev}} = \dfrac{2\pi \text{ rad}}{1.2 s}$

II) $v = r\omega = 2\left(\dfrac{2\pi}{1.2}\right) = \dfrac{4\pi}{1.2} = \boxed{10.47 \text{ m/s}}$

⑩ Find the area of each triangle below.
Round to the nearest hundredth.

i)

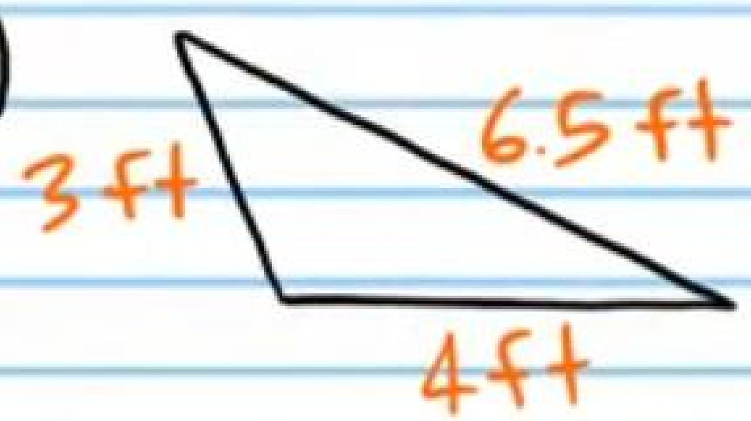


$\text{Area} = \frac{1}{2} bc \sin A$

$= \frac{1}{2}(10)(8)\sin 65°$

$\approx \boxed{36.25 \text{ cm}^2}$

ii)



$s = \dfrac{a+b+c}{2} = \dfrac{3+6.5+4}{2} = 6.75$

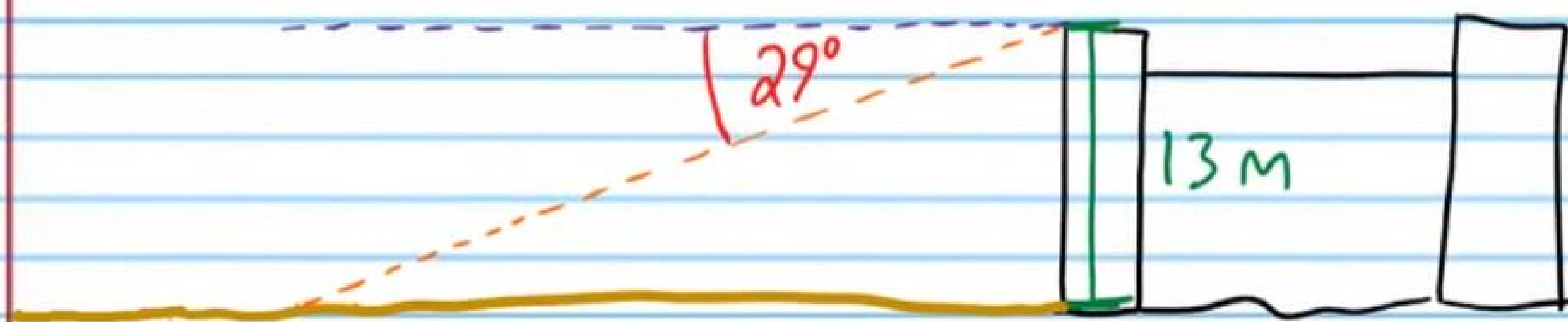
$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$

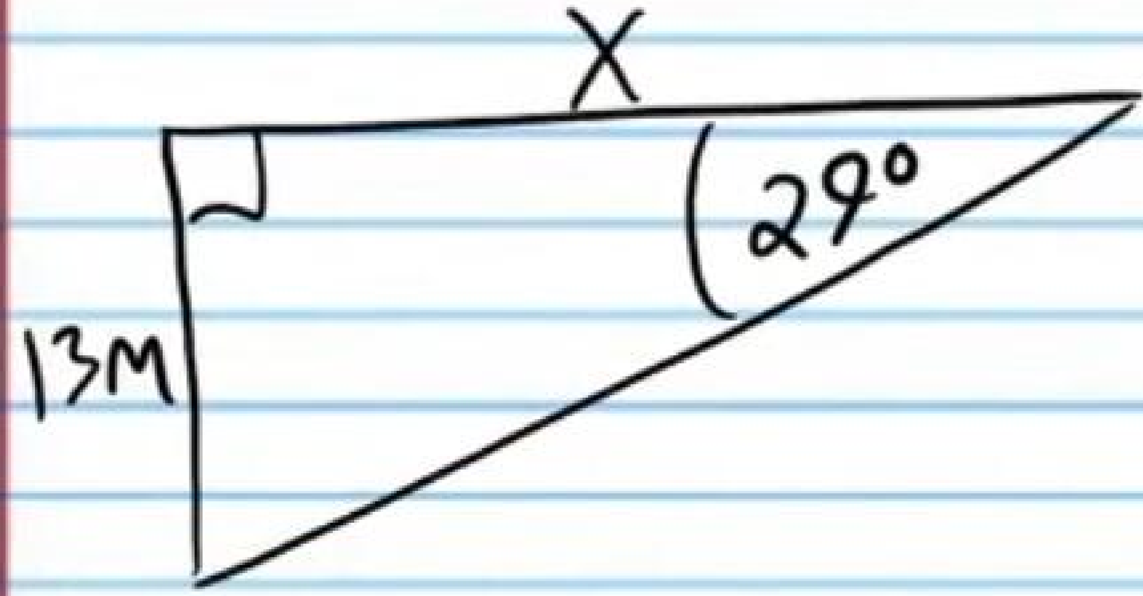
$= \sqrt{6.75(6.75-3)(6.75-6.5)(6.75-4)}$

$= \sqrt{6.75(3.75)(0.25)(2.75)}$

$= \boxed{4.17 \text{ ft}^2}$

(11) The angle of depression from the top of a castle to a point on the ground is 29°. If the castle is 13 meters high, how far is the point on the ground from the base of the castle? Round to the nearest hundredth.





$$\tan 29° = \frac{13}{X}$$

$$X \cdot \tan 29° = \frac{13}{X} \cdot X$$

$$X\tan 29° = 13$$

$$\frac{X\tan 29°}{\tan 29°} = \frac{13}{\tan 29°}$$

$$X = \frac{13}{\tan 29°}$$

$$X \approx \boxed{23.45\text{ M}}$$

(12) Find the values of the following expressions with a calculator. Round to the nearest ten thousandth.

i) $\cot 56° = \dfrac{1}{\tan 56°}$

$$\approx \boxed{0.6745}$$

ii) $\sec^{-1} 3$

$$\theta = \sec^{-1} 3$$

$$\sec\theta = 3$$

$$\frac{1}{\cos\theta} = 3$$

$$\cos\theta = \frac{1}{3}$$

$$\theta = \cos^{-1}\left(\frac{1}{3}\right)$$

$$\theta \approx \boxed{70.5288°}$$

(13) Find the value of the following expression without a calculator.

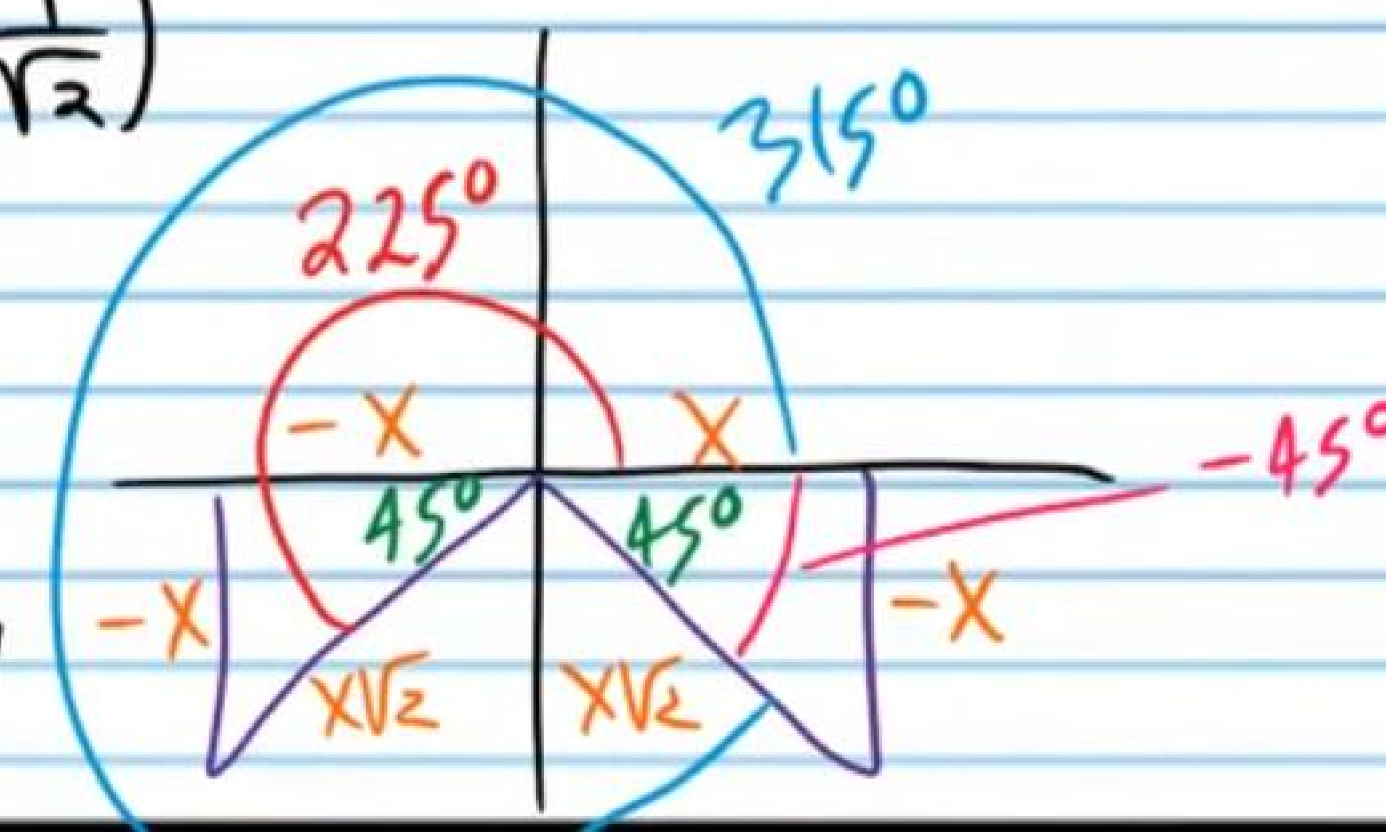
$$\arcsin\left(-\frac{1}{\sqrt{2}}\right) = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

$$\theta = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

$$\sin\theta = -\frac{1}{\sqrt{2}}$$

$$\theta = -45°$$

$$\arcsin\left(-\frac{1}{\sqrt{2}}\right) = \boxed{-45°}$$



(14) Solve if $0 < B < 2\pi$. Do not use a calculator.
Write your answers in radian units.

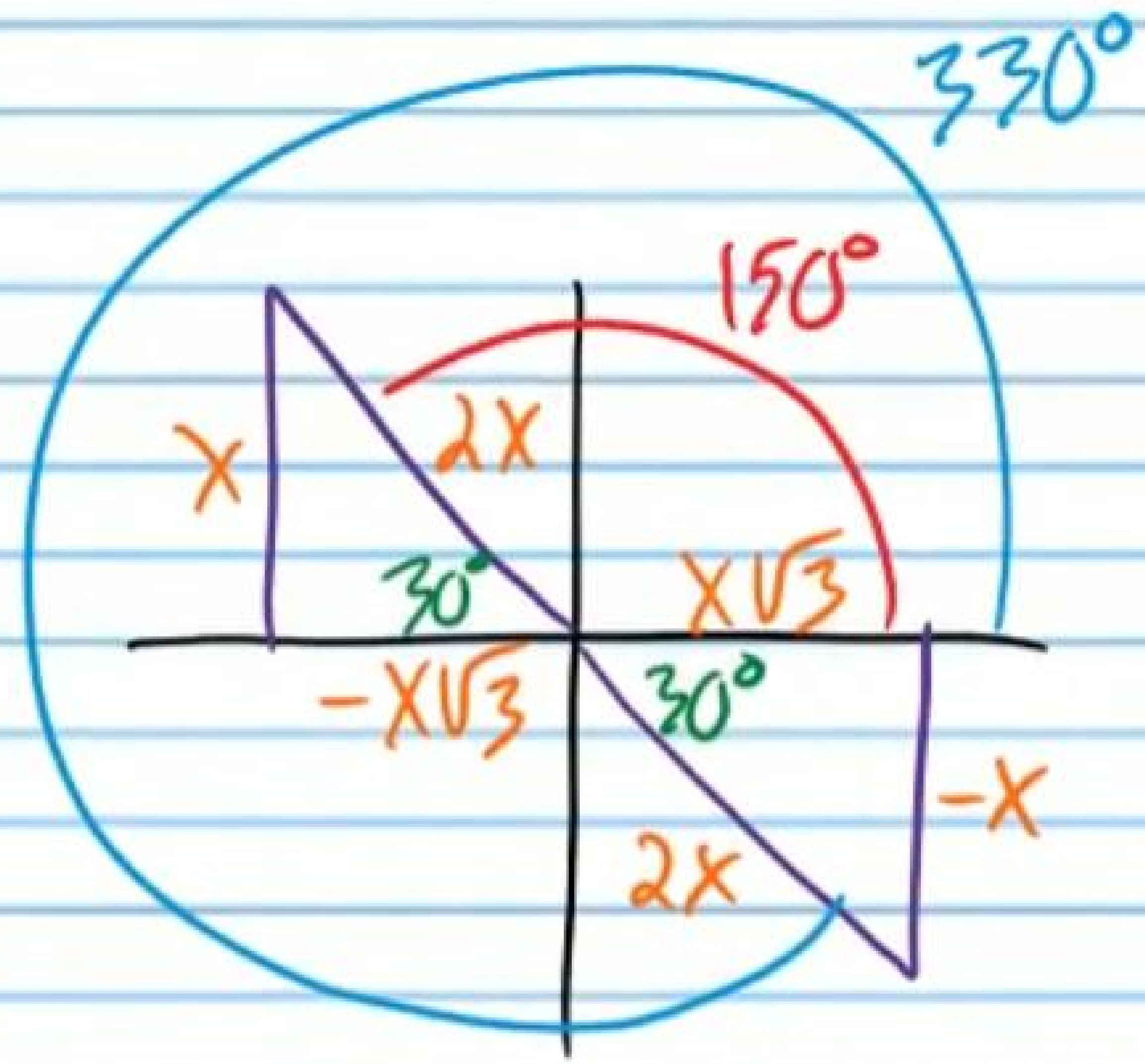
$$\cot B = -\frac{3}{\sqrt{3}}$$

$$\cot B = -\frac{3 \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}}$$

$$\cot B = -\frac{3\sqrt{3}}{\sqrt{9}}$$

$$\cot B = -\frac{3\sqrt{3}}{3}$$

$$\cot B = -\sqrt{3}$$



$$B = 150° \times \frac{\pi}{180°} = \frac{150\pi}{180} = \frac{15\pi}{18} = \boxed{\frac{5\pi}{6}}$$

$$B = 330° \times \frac{\pi}{180°} = \frac{330\pi}{180} = \frac{33\pi}{18} = \boxed{\frac{11\pi}{6}}$$

(15) Solve if $0 \le \beta < 360°$. Round to the nearest hundredth.

$$2\csc\beta - 5\csc\beta = -10$$

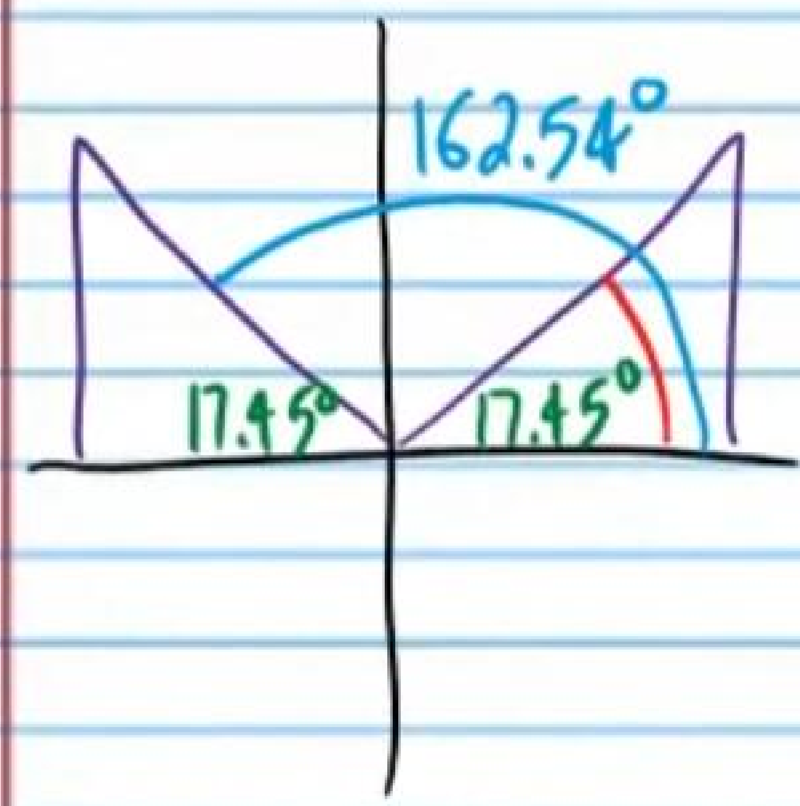
$$-3\csc\beta = -10$$

$$\csc\beta = \frac{10}{3}$$

$$\frac{1}{\sin\beta} = \frac{10}{3}$$

$$\sin\beta = \frac{3}{10}$$

$$\beta = \sin^{-1}\left(\frac{3}{10}\right)$$



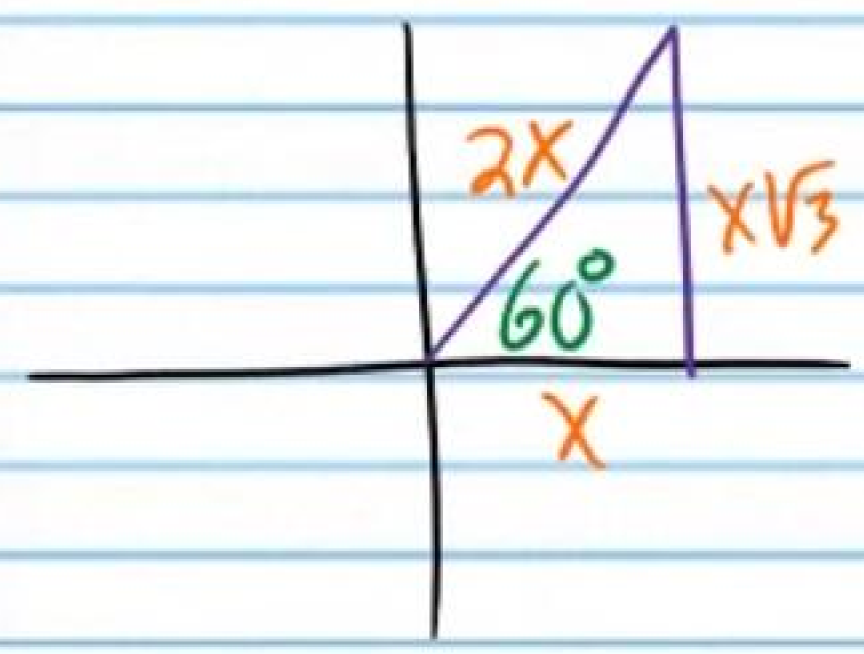
$$\boxed{\beta \approx 17.46°, 162.54°}$$

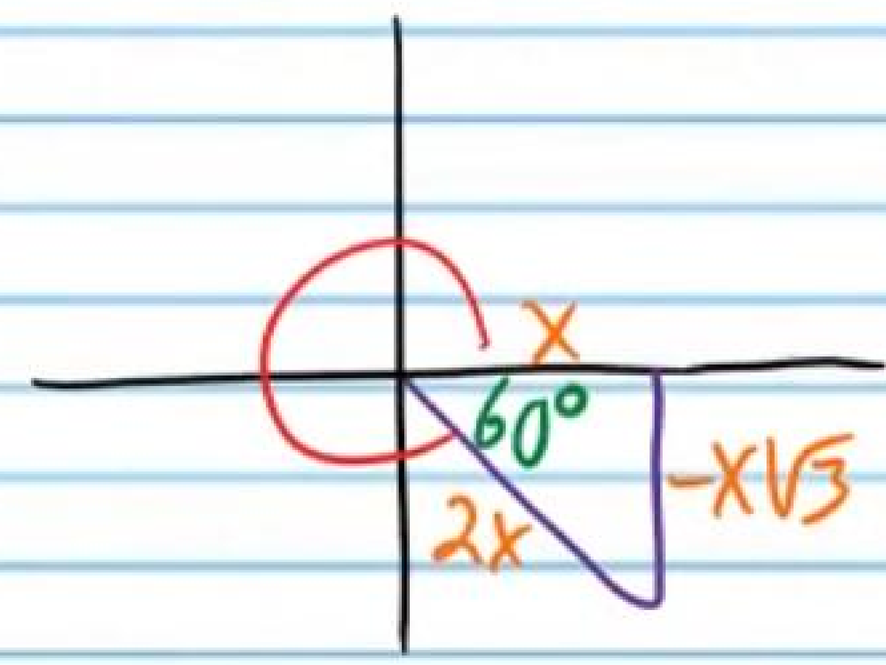
Find the value of each expression without a calculator.

(16) $\sin\left(\cos^{-1}\left(\frac{1}{2}\right)\right)$

I) $\theta = \cos^{-1}\left(\frac{1}{2}\right)$

$\cos\theta = \frac{1}{2}$





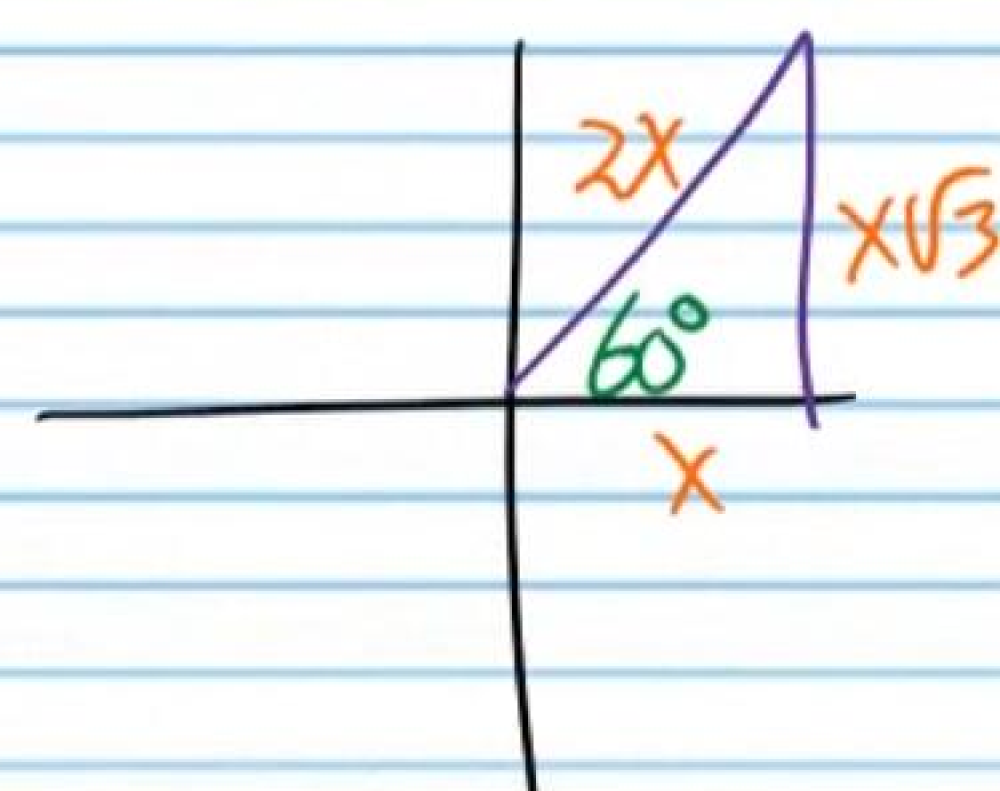
$\theta = 60°, 300°$

$\cos^{-1}\left(\frac{1}{2}\right) = 60°$

II) $\sin\left(\cos^{-1}\left(\frac{1}{2}\right)\right)$

$\sin 60°$

$\boxed{\frac{\sqrt{3}}{2}}$

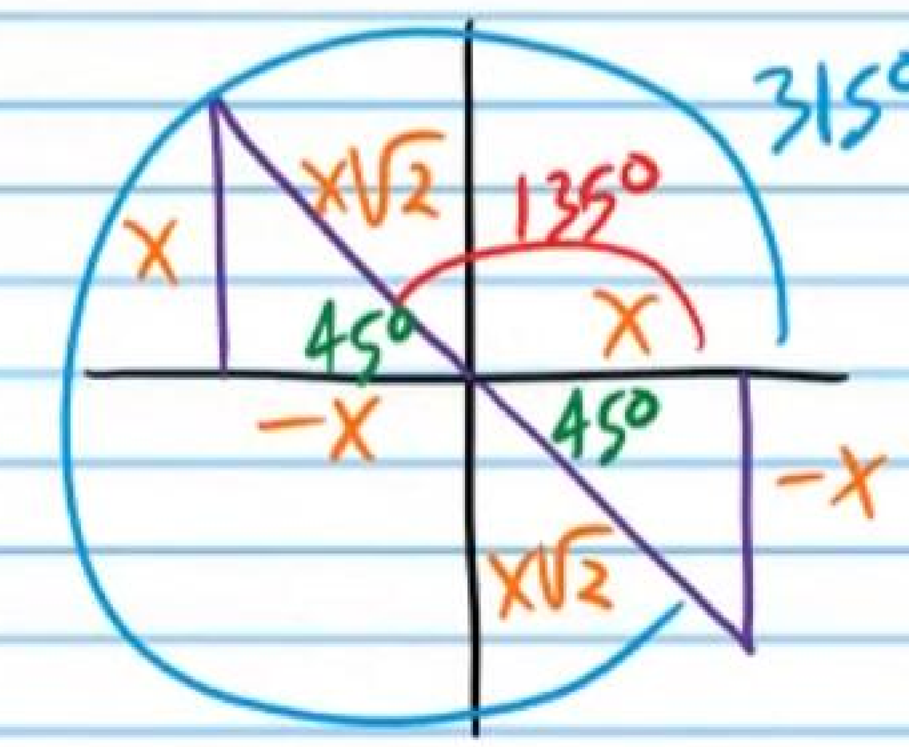


(17) $\csc\left(\cot^{-1}(-1)\right)$

I) $\theta = \cot^{-1}(-1)$

$\cot\theta = -1$

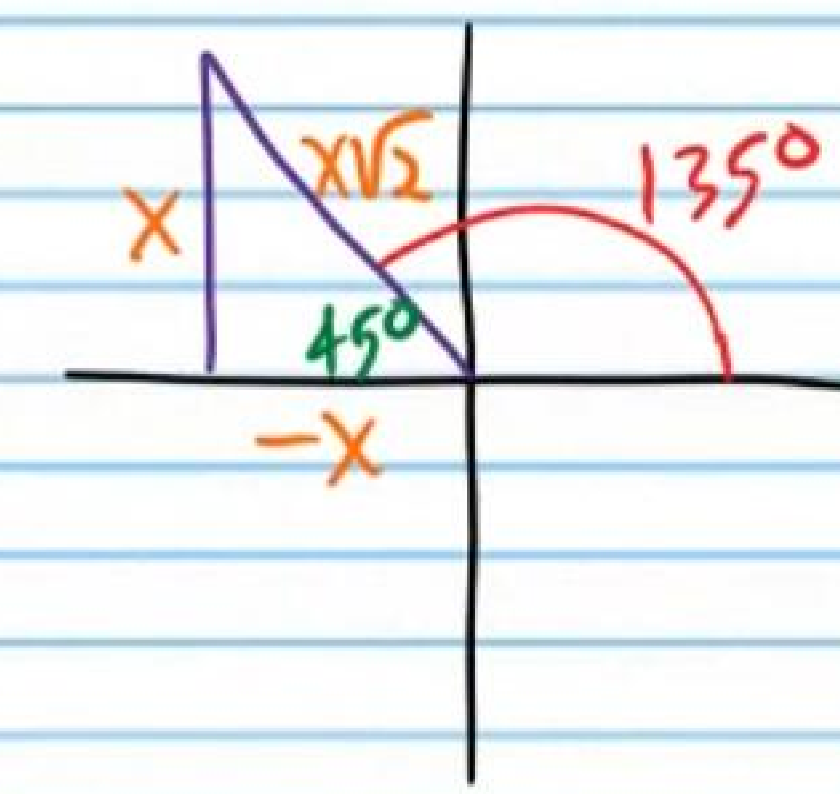
$\theta = 135°$



II) $\csc\left(\cot^{-1}(-1)\right)$

$\csc 135°$

$\boxed{\sqrt{2}}$



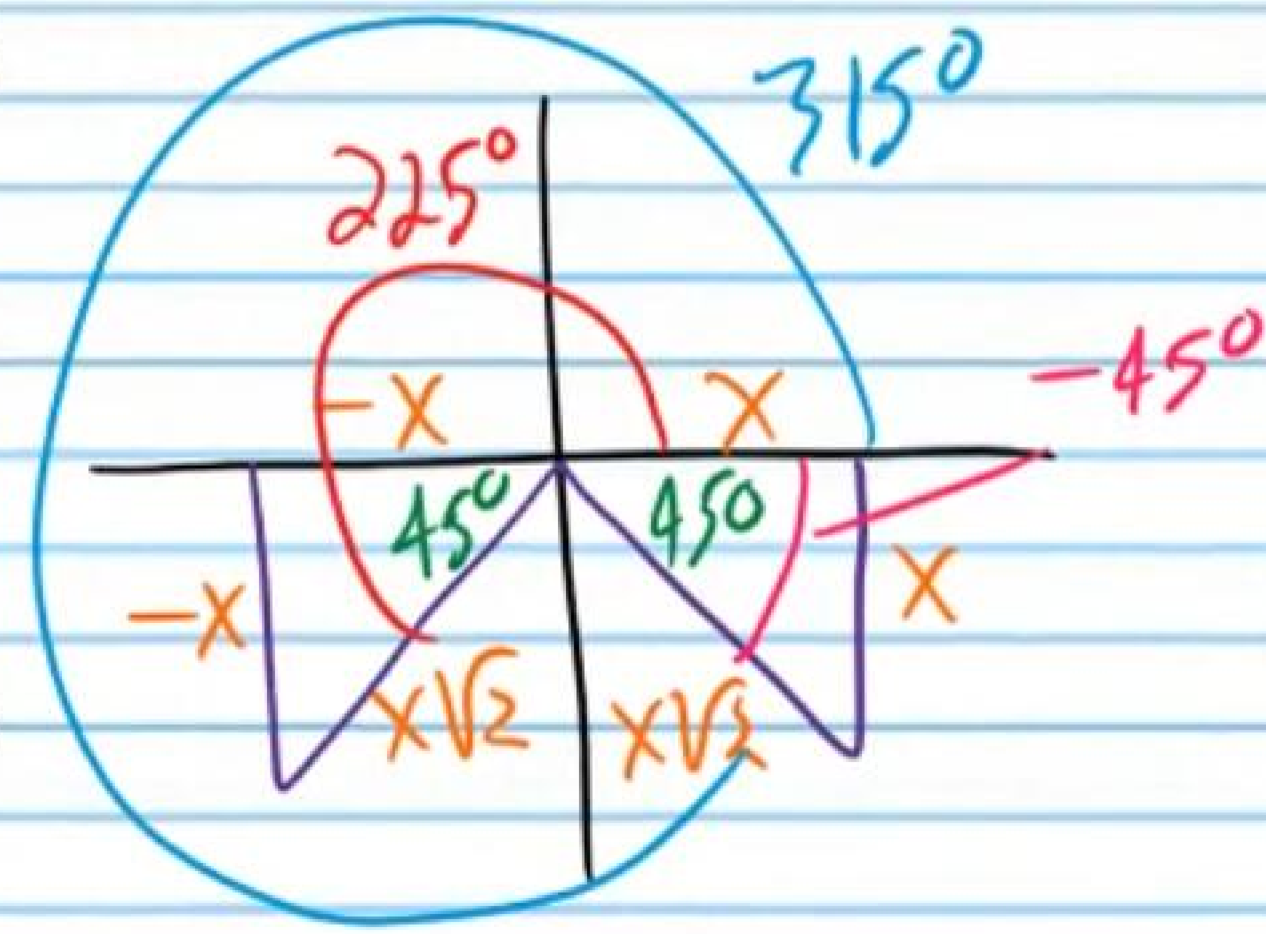
(18) $\tan(\csc^{-1}(-\sqrt{2}))$

I) $\theta = \csc^{-1}(-\sqrt{2})$

$\csc\theta = -\sqrt{2}$

$\theta = -45^\circ$

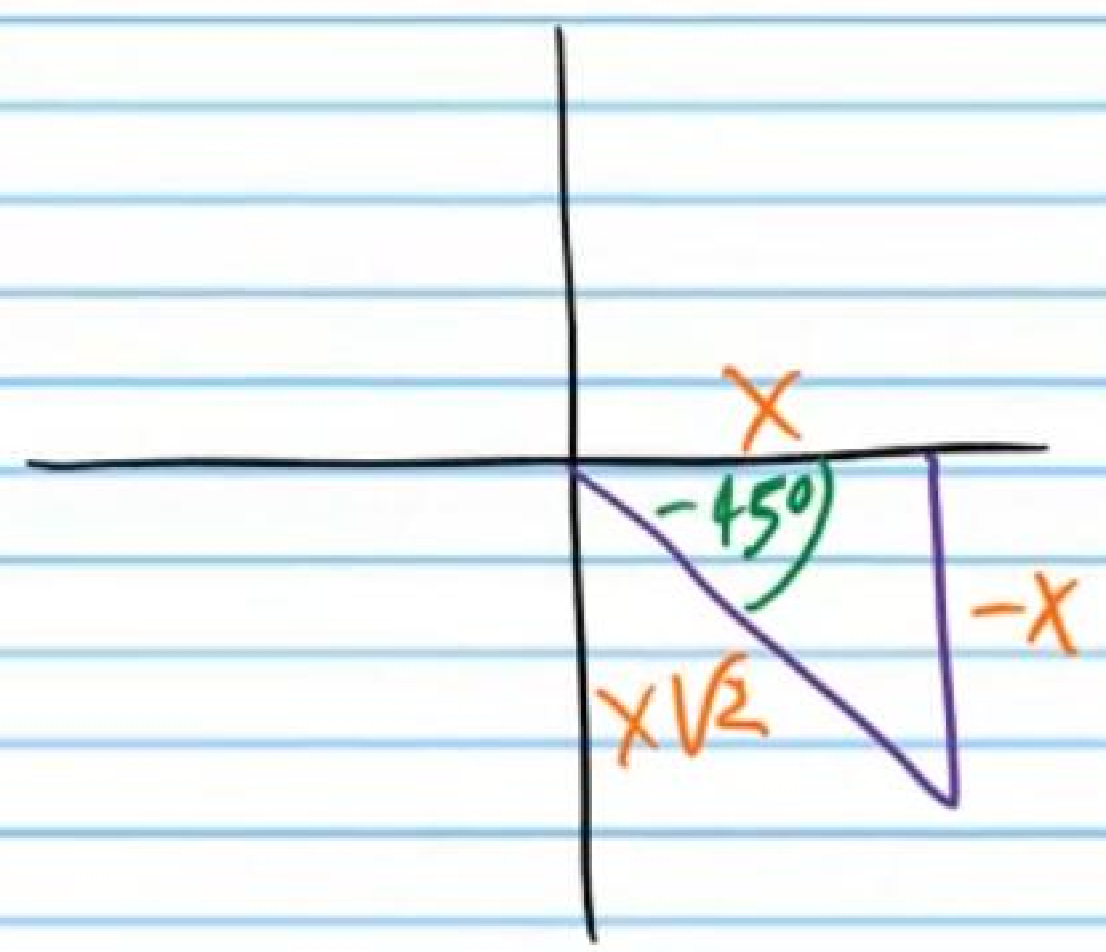
$\csc^{-1}(-\sqrt{2}) = -45^\circ$



II) $\tan(\csc^{-1}(-\sqrt{2}))$

$\tan(-45^\circ)$

$\boxed{-1}$



(19) $\sec(\sec^{-1}(\frac{2}{\sqrt{3}}))$

$\frac{2}{\sqrt{3}}$

$\boxed{\frac{2\sqrt{3}}{3}}$

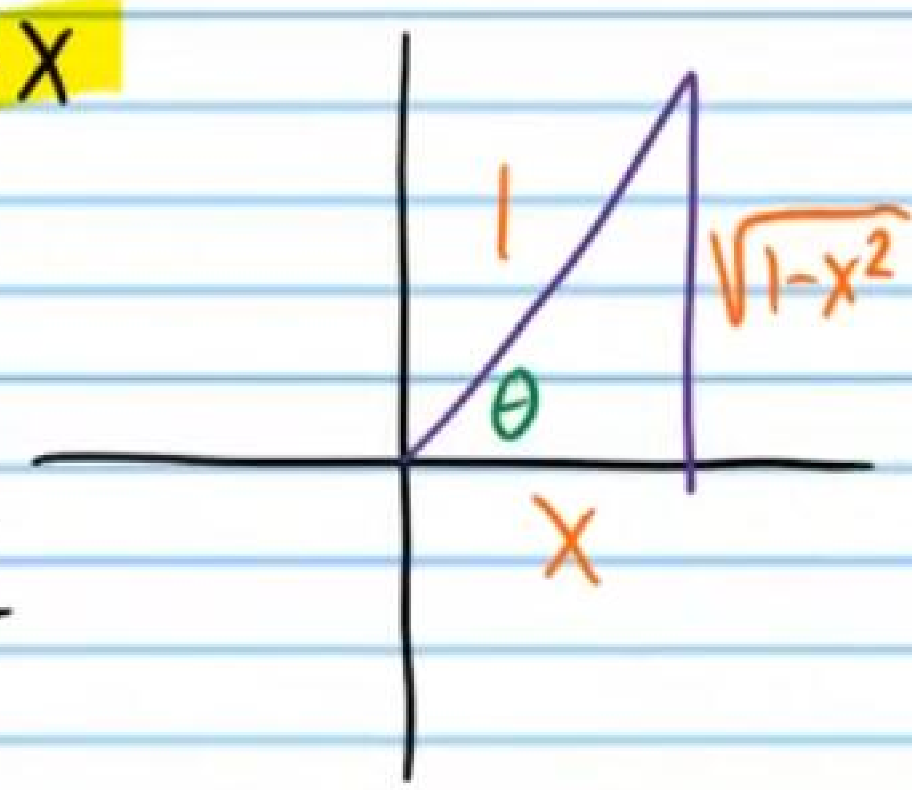
Write an equivalent expression without trigonometric notation. Assume that x is positive.

⑳ $\cot(\cos^{-1}x)$

I) $\theta = \cos^{-1}x$

$\cos\theta = x$

$\cos\theta = \frac{x}{1}$



$a^2+b^2=c^2$

$x^2+b^2=1^2$

$x^2+b^2=1$

$b^2=1-x^2$

$b=\pm\sqrt{1-x^2}$

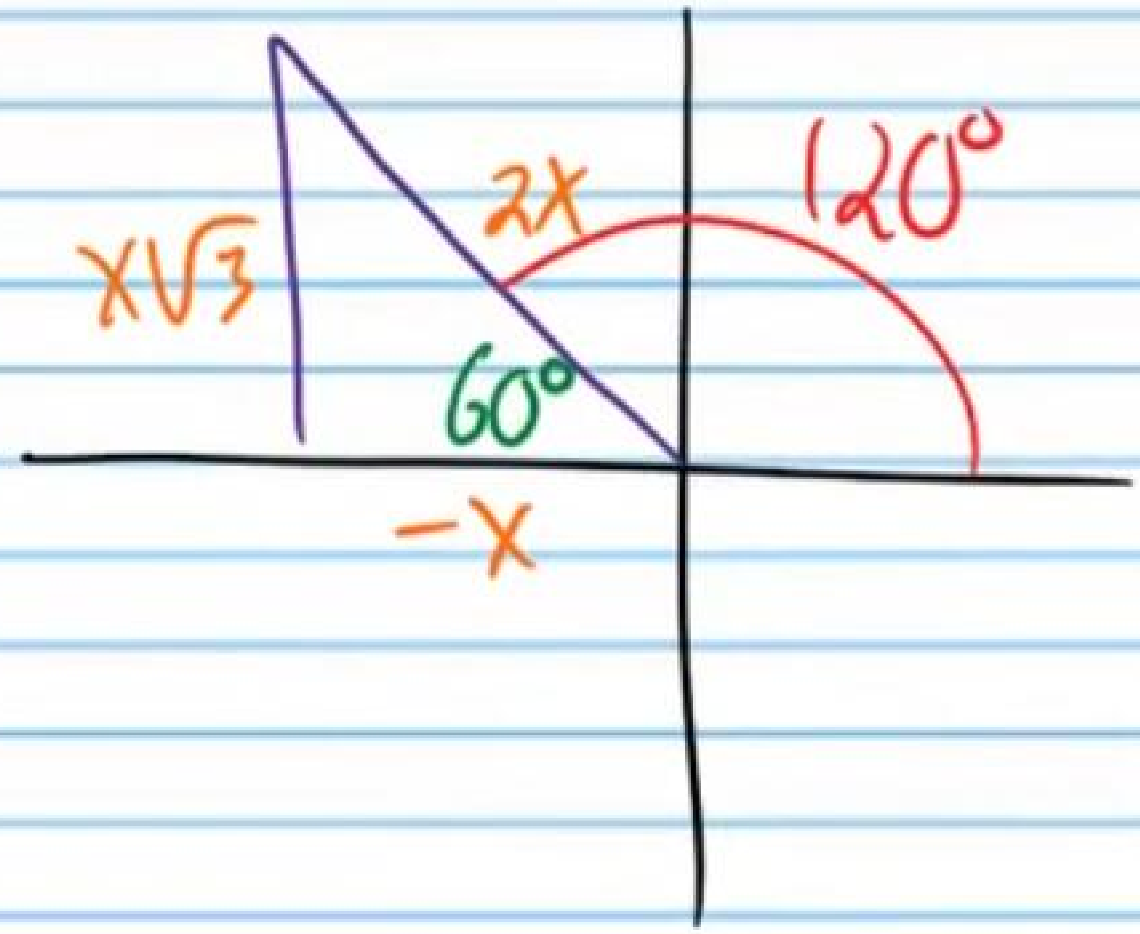
$b=\sqrt{1-x^2}$

II) $\cot(\cos^{-1}x) = \cot\theta = \boxed{\frac{x}{\sqrt{1-x^2}}}$

Find the value of each expression without a calculator.

㉑ $\tan^{-1}(\cot 120°)$

I) $\cot 120° = \frac{-1}{\sqrt{3}}$



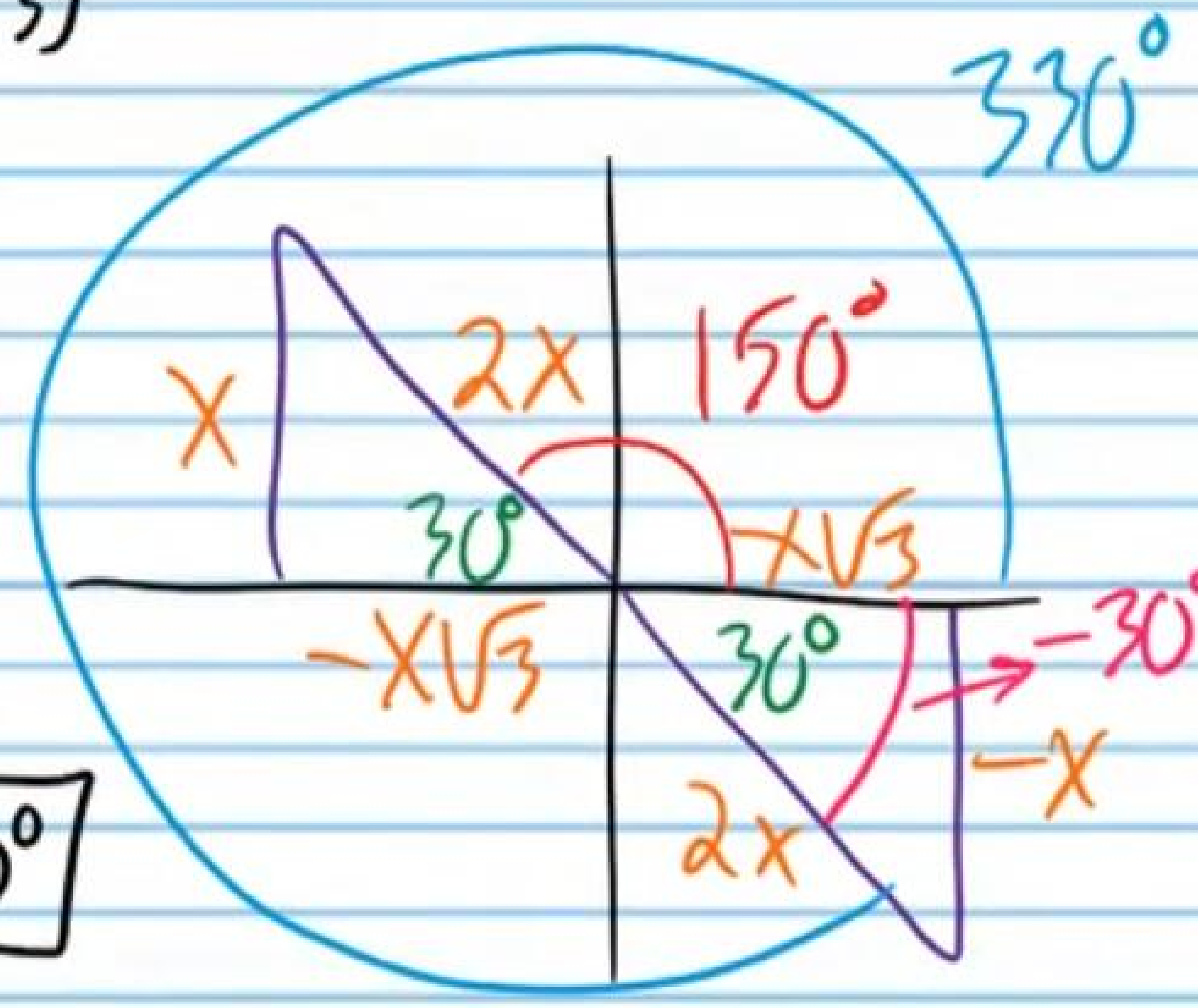
II) $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

$\theta = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

$\tan\theta = -\frac{1}{\sqrt{3}}$

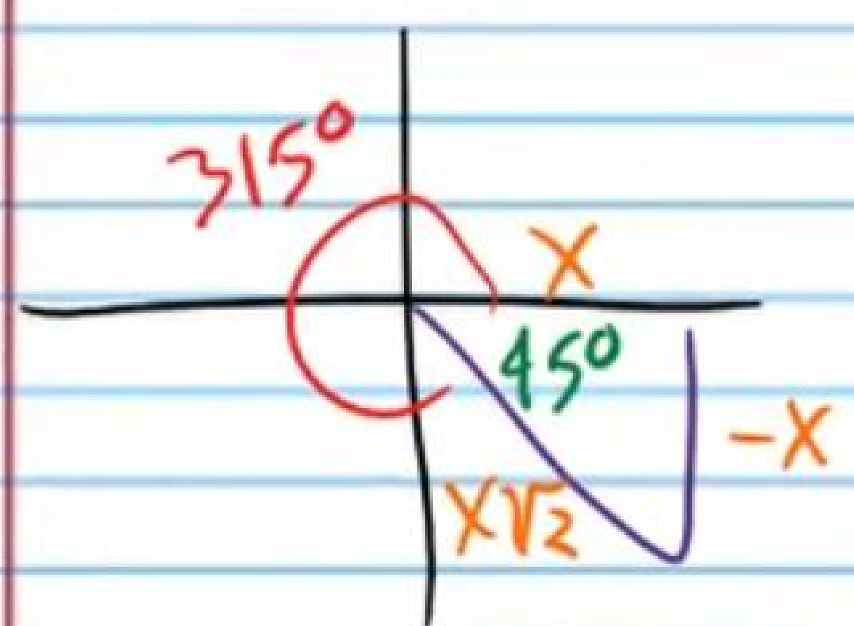
$\theta = -30°$

$\tan^{-1}(\cot 120°) = \boxed{-30°}$



22 i) $\sin^{-1}(\sin 315^\circ)$

I) $\sin 315^\circ$



$\sin 315^\circ = -\frac{1}{\sqrt{2}}$

II) $\sin^{-1}(\sin 315^\circ)$

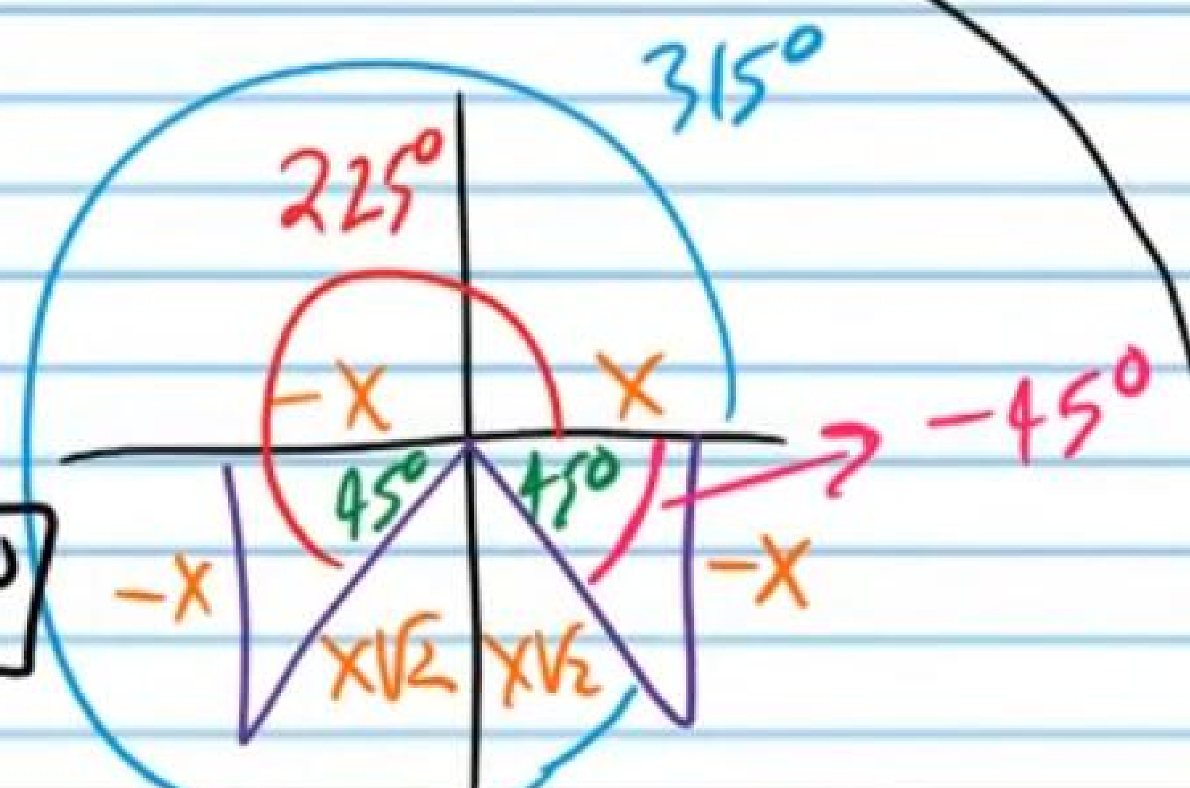
$\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

$\theta = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

$\sin\theta = -\frac{1}{\sqrt{2}}$

$\theta = -45^\circ$

$\sin^{-1}(\sin 315^\circ) = \boxed{-45^\circ}$



ii) $\cot^{-1}\left(\cot \frac{\pi}{2}\right)$

$\boxed{\frac{\pi}{2}}$

23 $\sec^{-1}\left(\csc \frac{2\pi}{3}\right)$

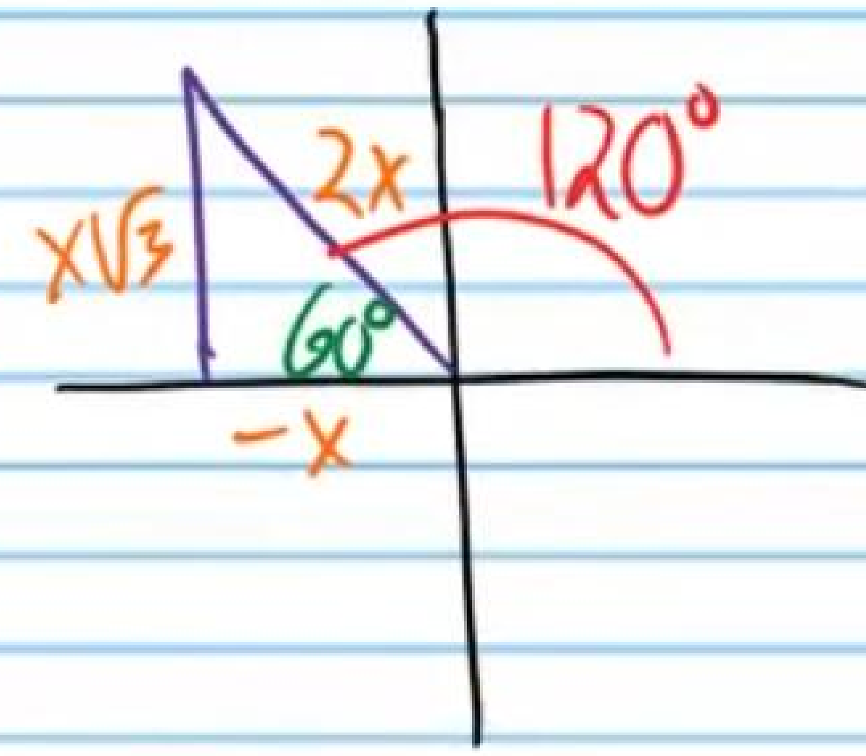
Conversion

$\frac{2\pi}{3} \times \frac{180^\circ}{\pi} = 120^\circ$

I) $\csc \frac{2\pi}{3}$

$\csc 120^\circ$

$\frac{2}{\sqrt{3}}$



II) $\sec^{-1}\left(\csc \frac{2\pi}{3}\right)$

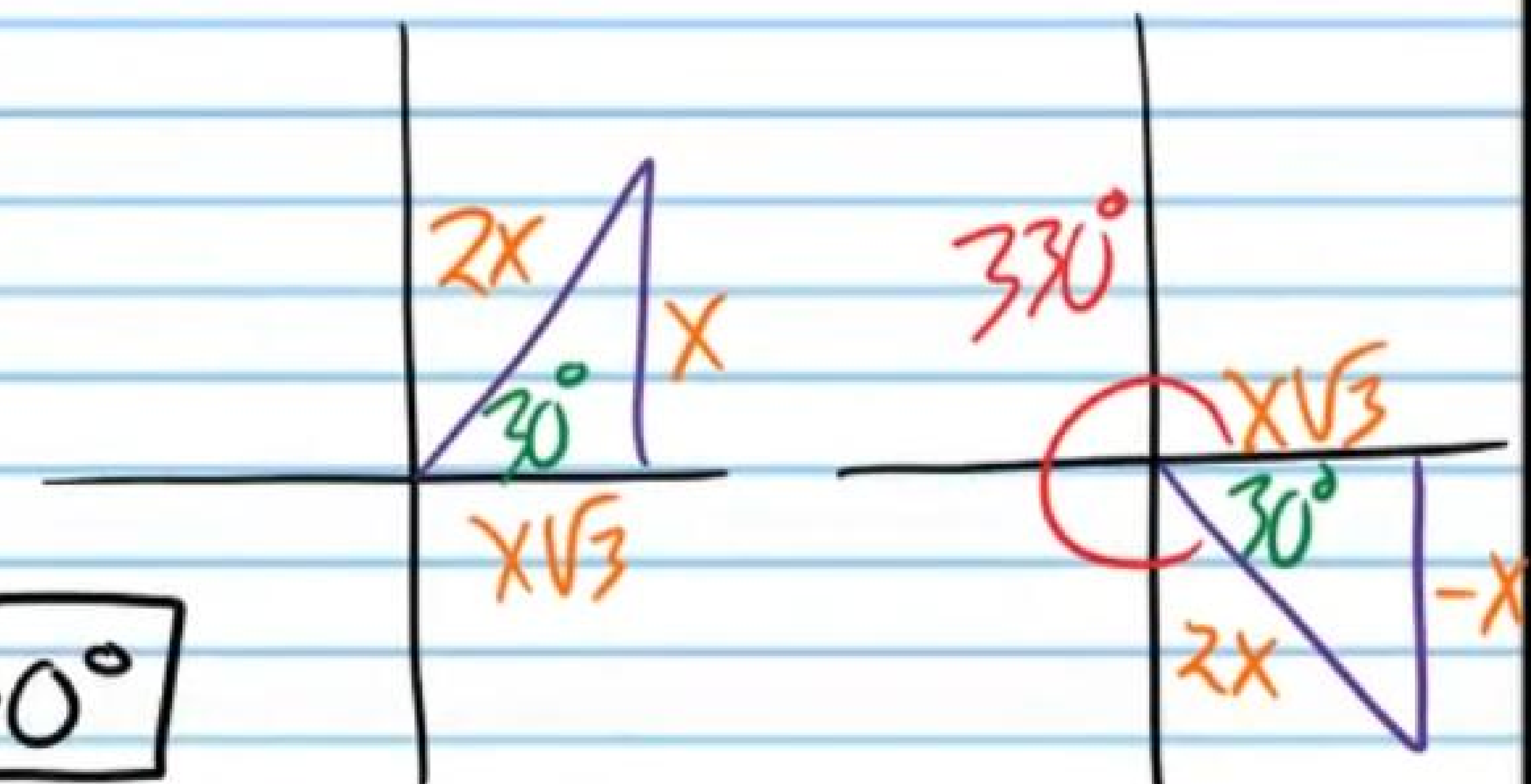
$\sec^{-1} \frac{2}{\sqrt{3}}$

$\theta = \sec^{-1} \frac{2}{\sqrt{3}}$

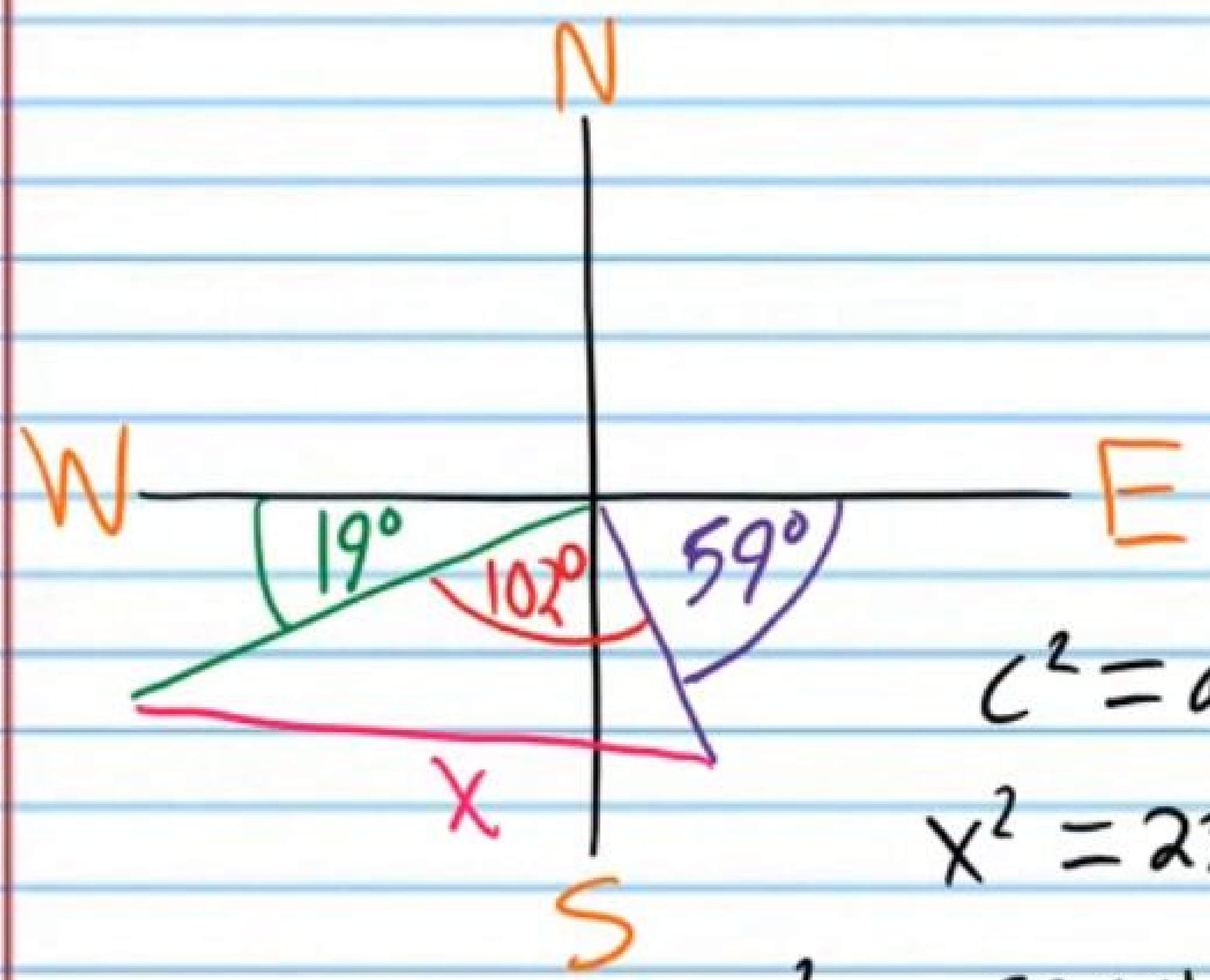
$\sec\theta = \frac{2}{\sqrt{3}}$

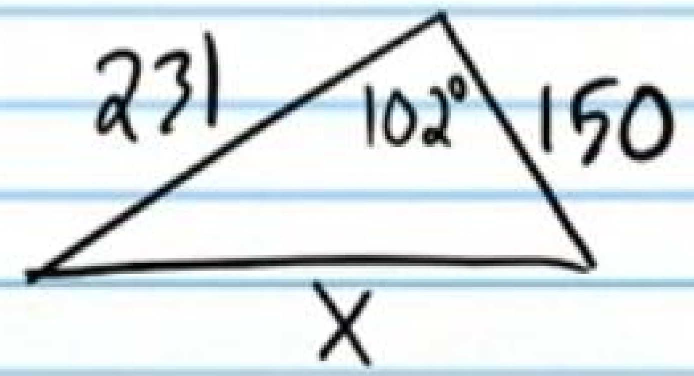
$\theta = 30^\circ$

$\sec^{-1}\left(\csc \frac{2\pi}{3}\right) = \boxed{30^\circ}$



(24) A gps device (global positioning system) shows that a plane must fly 231 kilometers on a bearing 19 degrees south of west (W19°S) to get to Oakdale Airport. The device also shows that the same plane must fly 150 kilometers on a bearing 59 degrees south of east (E59°S) to get to Harbor Airport. How far apart are the two airports?





$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$x^2 = 231^2 + 150^2 - 2(231)(150)\cos 102°$$

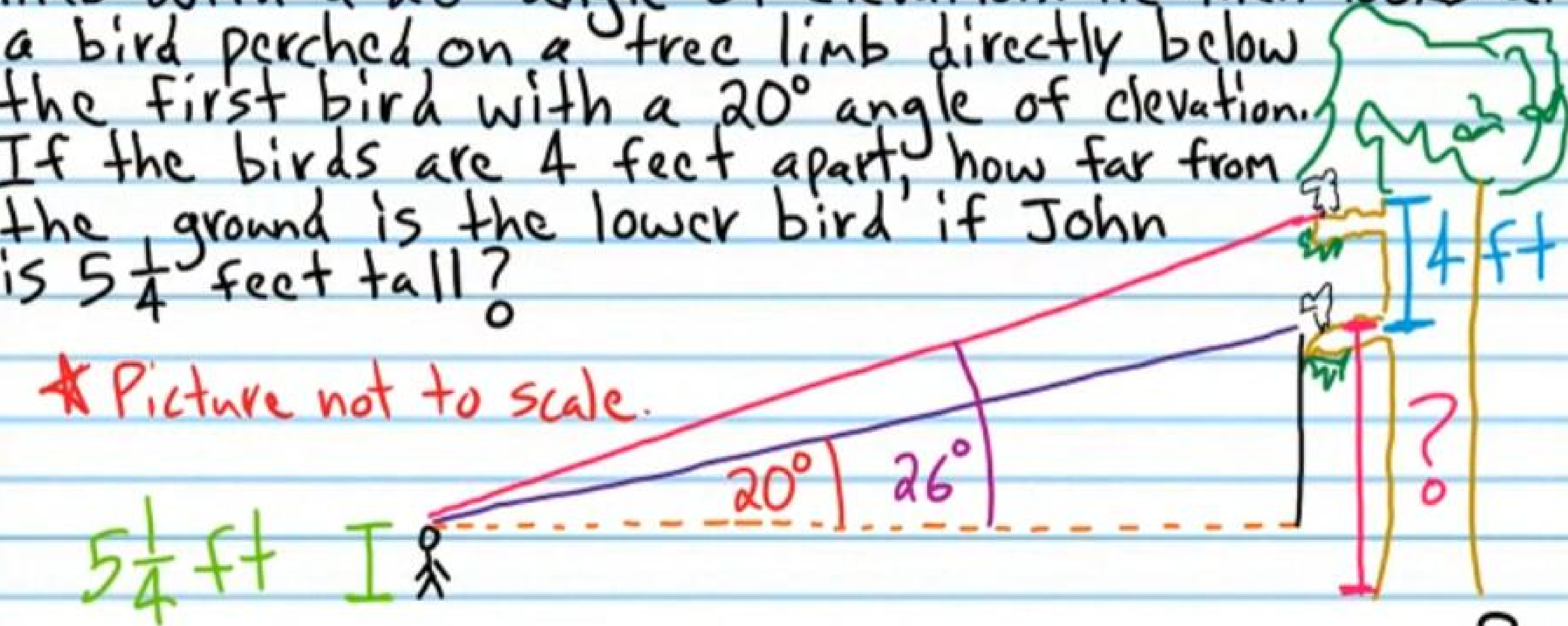
$$x^2 = 53{,}361 + 22{,}500 - 69{,}300\cos 102°$$

$$x^2 \approx 53{,}361 + 22{,}500 - (-14408.28)$$

$$x^2 \approx 90{,}269.28$$

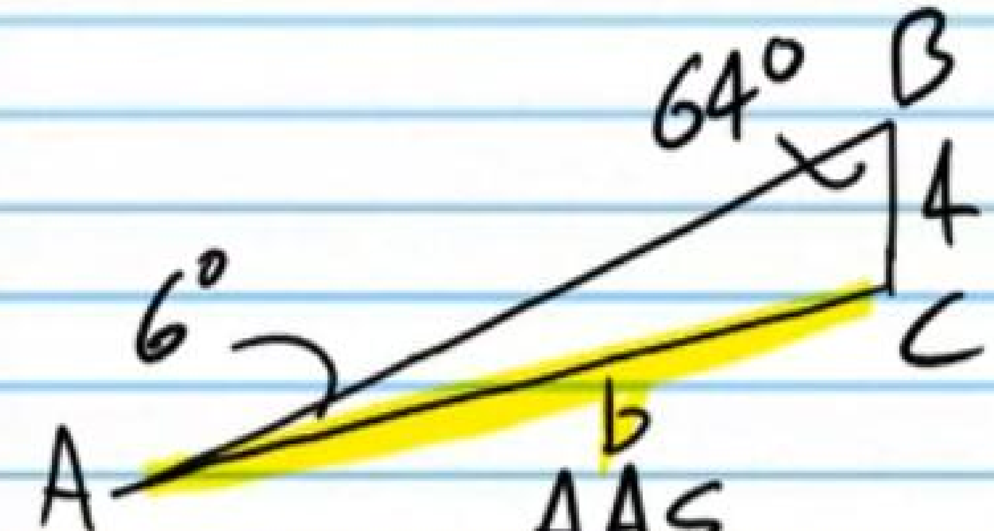
$$x \approx \boxed{300 \text{ KM}}$$

(25) John looks up at a bird perched on a tree limb with a 26° angle of elevation. He then looks at a bird perched on a tree limb directly below the first bird with a 20° angle of elevation. If the birds are 4 feet apart, how far from the ground is the lower bird if John is $5\frac{1}{4}$ feet tall?



I) $m\angle A = 26° - 20° = 6°$

$m\angle B = 180° - 26° - 90° = 64°$



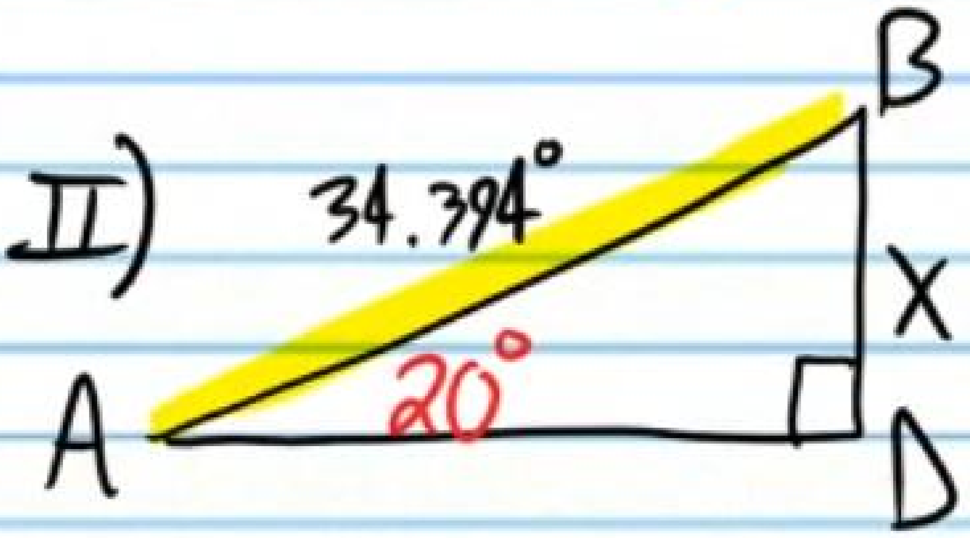
AAS

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 6°}{4} = \frac{\sin 64°}{b}$$

$$b = \frac{4\sin 64°}{\sin 6°} \approx 34.394...$$

II)



$$\sin 20° \approx \frac{x}{34.394}$$

$$x \approx 34.394\sin 20°$$

$$x \approx 11.7635...$$

III) $h \approx x + 5\frac{1}{4}$

$$h \approx 11.7635 + 5\frac{1}{4}$$

$$\boxed{h \approx 17.01 \text{ ft}}$$